THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 2: **Invisible Living World: Beyond Our Naked Eye** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A microscope has a 10x eyepiece and a 40x objective. A cell appears 2 mm wide in the image. The actual width is: 1

(a) 0.05 mm **(b)** 0.005 mm **(c)** 0.5 mm **(d)** 800 mm

2. A tissue differs from an organ because a tissue: 1

(a) has similar cells doing one job **(b)** forms systems

(c) has many types of tissues **(d)** is an organism

3. A cell is stained in blue and the nucleus appears dark. The role of stain is to: 1

(a) increase size **(b)** make food **(c)** kill the cell **(d)** provide contrast

4. A cell that has a nucleus, a cell wall and chloroplasts is likely from: 1

(a) a leaf **(b)** bacterium **(c)** human cheek **(d)** blood

5. A student gets a 400x total magnification using a 40x objective lens. The eyepiece lens is: 1

(a) 20x **(b)** 100x **(c)** 5x **(d)** 10x

6. A magnification of 100x means an object of 0.1 mm appears as: 1

(a) 0.01 mm **(b)** 1 mm **(c)** 10 mm **(d)** 100 mm

7. A student sees rectangular cells with a clear nucleus but no green bodies. The cells may be from: 1

(a) an onion bulb **(b)** a human cheek **(c)** a leaf **(d)** bacteria

8. Which of the following is the correct statement about plant cell vacuoles? 1

(a) They are large and store water and nutrients

(b) They are absent

(c) They are the control centre

(d) They make food

9. Which two are found in both a plant and an animal cell? 1

(a) wall and vacuole **(b)** chloroplast and wall

(c) nucleus and cytoplasm **(d)** chloroplast and nucleus only

10. Which of these statements about bacteria is correct? 1

(a) They are visible to the naked eye.

(b) They have genetic material but no true nucleus.

(c) They have a nucleus and chloroplasts.

(d) They are always harmful.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why are bacteria placed in a different group from plants and animals? 2

12. Differentiate between prokaryotic and eukaryotic cells (one point each). 2

13. Differentiate between cytoplasm and nucleus. 2

14. An eyepiece of 15x and an objective of 20x are used. Find the magnification. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A cell shows a thick wall, large vacuole and green bodies. Identify the cell, name the green bodies and state their function. 3

16. A cell is magnified 200x and measures 4 mm in the picture. Find its actual size in micrometres. 3

17. Name the cells A, B and C: (A) long, branched cell (B) disc-shaped without a nucleus (C) long and spindle-shaped that contracts. Give function of each. 3

OR

An unknown cell shows a nucleus but no wall or chloroplasts. Identify, with two reasons, the type of cell. 3

18. Name the organelles responsible for 3
(a) photosynthesis
(b) control
(c) storage in plant cells.

OR

Compare unicellular fungi and multicellular fungi with an example each. State one use of each. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Draw up a comparison between the plant cell, animal cell and bacterial cell in a table in terms of nucleus, cell wall, chloroplast, vacuole and size. 5

OR

What are the features of viruses? Mention two diseases caused by them and two ways of prevention. 5

20. Explain the significance of a microscope in biology. Give the parts, the working and the calculation of magnification for 12x eyepiece and 50x objective. 5

OR

Differentiate between prokaryotic and eukaryotic cells with respect to nucleus, size, examples and complexity (any four). 5

END OF QUESTION PAPER — Total: 40 Marks